

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
& MACHINE LEARNING**

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**AI POWERED STUDENT LEARNING
ASSISTANT
& PERSONALIZED CAREER AND
EDUCATION ADVISOR**

Bachelor of Engineering in Artificial Intelligence & Machine Learning

Submitted By:	Hari Priyan V
Department:	Artificial Intelligence & Machine Learning
College:	V.S.B Engineering college,karur.
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1. PROJECT OVERVIEW & BACKGROUND

The rapid evolution of Artificial Intelligence and Machine Learning has created unprecedented opportunities in the education sector. The **AI Powered Student Learning Assistant & Personalized Career and Education Advisor** is an integrated web-based platform designed to revolutionize how students learn, plan their careers, and navigate the complex landscape of modern professional requirements.

In today's fast-paced technological environment, students face multifaceted challenges: selecting the right career path, acquiring relevant skills, building effective resumes, and preparing for competitive job markets. Traditional educational systems often fail to provide personalized guidance tailored to individual student profiles, aspirations, and existing competencies. This project bridges that critical gap by leveraging state-of-the-art Generative AI, Natural Language Processing, and intelligent recommendation systems.

1.1 Background Context

According to a 2025 UNESCO report, approximately 60% of engineering graduates worldwide feel underprepared for industry requirements. A National Employability Report indicates that only 45% of engineering graduates are immediately employable, primarily due to skill mismatches and lack of structured career guidance. The Indian education system graduates over 1.5 million engineering students annually, yet a significant proportion struggle to translate academic knowledge into professional competence.

This platform addresses these challenges by providing an AI-driven ecosystem that integrates personalized learning roadmaps, real-time career counseling, resume analysis, mock interviews, and internship recommendations — all tailored to each student's unique profile.

1.2 Project Significance

Dimension	Impact
Personalization	AI tailors content to individual student learning pace, style, and goals
Career Alignment	Bridges gap between academic skills and industry requirements
Accessibility	24/7 AI guidance accessible to students regardless of geography
Data-Driven	Analytics-powered insights for continuous improvement
Industry-Ready	Mock interviews and skill assessments aligned to real job descriptions

2. PROBLEM IDENTIFICATION

A comprehensive analysis of the current educational landscape reveals several critical pain points that adversely affect student academic performance and career readiness. These problems have been identified through literature review, student surveys, and industry feedback.

2.1 Core Problems

1. Lack of Personalized Learning

Students receive uniform content regardless of their learning speed, prior knowledge, or preferred learning style. One-size-fits-all curricula fail to address individual competency gaps.

2. Absence of Career Guidance

Most educational institutions lack dedicated career counselors with up-to-date industry knowledge. Students make career decisions based on incomplete or outdated information.

3. Poor Resume Quality

Students are unaware of industry standards for resumes. ATS (Applicant Tracking System) optimization is rarely taught, resulting in qualified candidates being filtered out.

4. Interview Unpreparedness

Students lack access to realistic mock interview environments. Fear of interviews and lack of practice significantly reduce placement rates.

5. Skill Gap Unawareness

Students are often unaware of the specific skills required for their desired roles. The gap between academic learning and industry expectations remains unaddressed.

6. Limited Internship Access

Finding relevant internships is a time-consuming, unstructured process. Students lack intelligent matching with opportunities aligned to their profiles and goals.

7. Absence of Learning Analytics

Students have no mechanism to track their learning progress quantitatively. Educators lack data-driven insights to improve curriculum effectiveness.

8. Information Overload

The abundance of online learning resources creates decision paralysis. Students cannot identify which resources are most relevant to their specific goals.

2.2 Problem Impact Analysis

Problem	Severity	Root Cause	Proposed Solution
Low Placement Rate	High	Skill mismatch	AI skill assessment & roadmap
Resume Rejections	High	ATS incompatibility	AI resume analyzer
Interview Failures	High	No practice platform	Mock interview system

Problem	Severity	Root Cause	Proposed Solution
Career Confusion	Medium	No guidance system	AI career advisor
Missed Internships	Medium	No smart matching	Recommendation engine
Learning Inefficiency	High	Generic content	Personalized roadmap

3. PROJECT OBJECTIVES & SUCCESS CRITERIA

3.1 Primary Objectives

- **Objective 1:** Develop an AI-powered personalized learning assistant that adapts to individual student profiles, learning styles, and knowledge levels.
- **Objective 2:** Create an intelligent career advisor module capable of providing data-driven career pathway recommendations based on student skills, interests, and market trends.
- **Objective 3:** Build a comprehensive resume analyzer that evaluates resume quality, provides ATS optimization suggestions, and generates improvement recommendations.
- **Objective 4:** Implement a realistic mock interview system with AI-generated domain-specific questions and intelligent feedback mechanisms.
- **Objective 5:** Design a skill gap analysis engine that identifies discrepancies between current student skills and job role requirements.
- **Objective 6:** Develop an internship recommendation system using collaborative filtering and content-based algorithms.
- **Objective 7:** Create a learning analytics dashboard providing visual insights into student progress, strengths, and improvement areas.
- **Objective 8:** Ensure scalable, secure, and performant system architecture capable of supporting concurrent multi-user interactions.

3.2 Success Criteria

ID	Criterion	Target Metric	Measurement Method
SC-01	AI Response Accuracy	>= 85% relevance score on learning recommendations	User satisfaction survey + AI evaluation
SC-02	Resume Analysis Speed	< 10 seconds per resume	System performance testing
SC-03	System Uptime	>= 99% availability	Infrastructure monitoring
SC-04	User Adoption	>= 200 active student users within 30 days	User registration analytics
SC-05	Interview Feedback	AI feedback within 30 seconds	Response time logging
SC-06	Skill Gap Accuracy	90% alignment with industry job descriptions	Cross-validation with job portals
SC-07	API Response Time	< 2 seconds for 95% of requests	Load testing with JMeter
SC-08	Security Compliance	Zero critical vulnerabilities	Penetration testing

4. PROJECT SCOPE & BOUNDARIES

4.1 In Scope

- Web-based application accessible via modern browsers (Chrome, Firefox, Edge, Safari)
- Student registration, authentication, and profile management
- AI-powered conversational learning assistant using Gemini/OpenAI API
- Personalized learning roadmap generation based on student goals and skills
- Resume upload, parsing, and AI-driven analysis with improvement suggestions
- Mock interview simulator with domain-specific question generation and evaluation
- Skill gap analysis comparing student profile against job role requirements
- Internship recommendation engine with filtering by location, domain, and duration
- Learning analytics dashboard with charts and progress visualization
- Progress tracking with milestone completions and achievement badges
- JWT-based secure authentication with role-based access control
- MySQL relational database for persistent data storage
- RESTful API backend using Flask Python
- Responsive frontend using React.js and Tailwind CSS

4.2 Out of Scope

- Mobile native application development (iOS/Android) — web-responsive only
- Payment gateway integration for premium features
- Video conferencing or live mentoring sessions
- Automated job application submission on behalf of students
- Multi-language support (English only in initial version)
- Integration with university ERP or LMS systems
- Proctoring or anti-cheating mechanisms for assessments
- Social networking or peer-to-peer communication features

4.3 System Boundaries

Boundary Type	Description
Input Boundary	Student profile data, resume files (PDF/DOCX), text queries, quiz responses, skill ratings
Processing Boundary	AI API calls, database operations, recommendation algorithms, analytics computations
Output Boundary	Learning roadmaps, career recommendations, resume feedback, interview Q&A;, internship listings, analytics reports
External APIs	Gemini API / OpenAI GPT-4, job listing APIs (optional), email notification service
Data Boundary	MySQL database on local/cloud server; no external data sharing without consent

5. USER ROLES & STAKEHOLDER ANALYSIS

5.1 User Roles

Student (Primary user)

- Register and create a profile
- Interact with AI learning assistant
- Generate and follow personalized roadmaps
- Upload resume for AI analysis
- Participate in mock interviews
- View skill gap analysis and recommendations
- Track learning progress and analytics
- Browse and save internship opportunities

Administrator (System manager)

- Manage user accounts and permissions
- Monitor system health and performance
- Configure AI model parameters
- Manage internship listing database
- Generate system-wide analytics reports
- Handle content moderation and quality

AI System (Automated agent)

- Process natural language queries
- Generate personalized recommendations
- Analyze resume content and provide feedback
- Generate interview questions and evaluate responses
- Compute skill gap metrics
- Update and adapt learning roadmaps

5.2 Stakeholder Analysis

Stakeholder	Involvement	Interest	Expected Benefit
Students	High	High	Improved career readiness and placement success
Faculty/Professors	Medium	Medium	Curriculum insights; reduced repetitive mentoring load
Placement Officers	Medium	High	Higher placement rates; data-driven student profiles
College Management	Low	High	Institutional reputation improvement; accreditation support
Recruiters	Low	High	Access to better-prepared, ATS-optimized candidates
AI API Providers	High	Medium	API usage and revenue generation
Developers	High	Low	System functionality and maintenance

6. EXISTING SYSTEM

6.1 Overview of Current Solutions

Several platforms exist that partially address student learning and career development needs. However, none provides a fully integrated, AI-powered solution that unifies learning, career guidance, resume analysis, and interview preparation within a single cohesive ecosystem.

Platform	Primary Feature	Limitations
LinkedIn Learning	Video-based courses	No personalized roadmap; No career advice; No resume analyzer; Subscription-based cost barrier
Coursera / edX	MOOCs and certifications	Generic course recommendations; No skill gap analysis; No interview prep; No internship matching
Resume.io	Resume builder and templates	No AI analysis; No career counseling; No learning integration; Template-only approach
Glassdoor / Naukri	Job listings and company reviews	No learning features; No personalized guidance; Only job browsing; No skill development
InterviewBit	Coding interview preparation	Limited to technical roles; No resume analysis; No career advisor; Domain-specific only
ChatGPT (Generic)	General AI assistance	No student-specific features; No profile tracking; No analytics; No structured roadmap

6.2 Limitations of Existing Systems

- Fragmented ecosystem requiring students to use multiple separate platforms
- No unified student profile that persists across learning and career modules
- Lack of real-time personalization based on individual skill assessments
- No integration between resume quality and learning recommendations
- Generic AI responses without domain-specific academic context
- High subscription costs creating access barriers for students
- No progress tracking linked to career readiness metrics
- Absence of intelligent internship matching based on student profiles

7. PROPOSED SYSTEM

7.1 System Description

The proposed system is a fully integrated AI-powered platform that serves as a one-stop solution for student learning, skill development, and career preparation. The system leverages Google's Gemini API (or OpenAI GPT-4) as its AI backbone, providing intelligent, context-aware responses tailored to each student's unique profile. Built on a robust React.js frontend, Flask Python backend, and MySQL database, the system ensures high performance, scalability, and security.

7.2 Key Advantages Over Existing Systems

Feature	Existing Systems	Proposed System
Integration	Fragmented platforms	Unified ecosystem with 10+ features
Personalization	Generic recommendations	AI-driven individual profile-based guidance
AI Quality	No AI or basic chatbots	State-of-the-art LLM (Gemini/GPT-4)
Career Guidance	Job listings only	Comprehensive career advisor with roadmaps
Analytics	No learning analytics	Visual dashboard with progress metrics
Interview Prep	Limited or domain-specific	Multi-domain AI mock interviews
Cost	High subscription fees	Accessible, institution-deployable solution
Resume AI	Template-only builders	Deep AI analysis with ATS optimization

7.3 System Modules Overview

ID	Module	Description
M1	AI Learning Assistant	Conversational AI for subject queries, concept explanations, and personalized Q&A;
M2	Personalized Roadmap	AI-generated step-by-step learning paths based on goal, current skills, and timeline
M3	Career Advisor	AI-powered career counseling with role recommendations and industry insights
M4	Resume Analyzer	PDF/DOCX resume parsing with AI feedback, ATS scoring, and improvement tips
M5	Skill Gap Analysis	Comparison of student skills vs job role requirements with gap identification
M6	Mock Interview System	Domain-specific question generation, response evaluation, and feedback
M7	Internship Recommender	Collaborative filtering-based internship matching with profile alignment
M8	Learning Analytics Dashboard	Visual charts for progress, time spent, skill growth, and career readiness
M9	Student Profile Management	Comprehensive profile with academic background, skills, and career goals
M10	Progress Tracking	Milestone tracking, achievement badges, and completion certificates

8. FUNCTIONAL REQUIREMENTS

Functional requirements define the specific behaviors, functions, and capabilities the system must provide to its users. Each requirement is assigned a unique identifier, priority level, and description.

ID	Feature	Priority	Description
FR-001	User Registration	High	System shall allow students to register with email, name, department, year, and password
FR-002	User Authentication	High	System shall authenticate users using JWT tokens with 24-hour expiry
FR-003	Profile Management	High	System shall allow students to create and update profiles with skills, interests, and career goals
FR-004	AI Chat Assistant	High	System shall provide an AI conversational interface for academic queries using LLM API
FR-005	Roadmap Generation	High	System shall generate personalized learning roadmaps based on student goals and current skills
FR-006	Career Advice	High	System shall provide AI-generated career pathway recommendations with role descriptions
FR-007	Resume Upload	High	System shall accept resume uploads in PDF/DOCX formats up to 5MB
FR-008	Resume Analysis	High	System shall analyze resumes and return ATS score, strengths, weaknesses, and suggestions
FR-009	Skill Gap Analysis	High	System shall compare student skills with job role requirements and identify gaps
FR-010	Mock Interview	High	System shall generate interview questions by domain and evaluate AI answers with feedback
FR-011	Internship Search	Medium	System shall display internship listings with filters for domain, location, and duration
FR-012	Internship Match	Medium	System shall recommend internships based on student profile using matching algorithms
FR-013	Progress Tracking	High	System shall track completion status of roadmap modules and display percentage completion
FR-014	Analytics Dashboard	Medium	System shall display charts for skills learned, time spent, and career readiness score
FR-015	Badge System	Low	System shall award achievement badges upon milestone completions
FR-016	Admin Dashboard	Medium	Administrator shall be able to view user statistics, manage content, and monitor system health
FR-017	Search Functionality	Medium	System shall provide a global search across topics, roadmaps, and internship listings
FR-018	Notification System	Low	System shall send email notifications for profile completions and new internship matches

9. NON-FUNCTIONAL REQUIREMENTS

ID	Category	Requirement
NFR-001	Performance	API response time < 2s for 95% requests; Page load < 3s
NFR-002	Scalability	Support 500 concurrent users without performance degradation
NFR-003	Availability	99% uptime during academic hours (8 AM – 10 PM IST)
NFR-004	Security	HTTPS, JWT authentication, bcrypt password hashing, SQL injection prevention
NFR-005	Data Privacy	Student data stored encrypted; No third-party data sharing without consent
NFR-006	Usability	Intuitive UI with System Usability Scale (SUS) score >= 75
NFR-007	Reliability	Mean Time Between Failures (MTBF) >= 720 hours
NFR-008	Maintainability	Modular codebase with code coverage >= 70%; documented APIs
NFR-009	Compatibility	Compatible with Chrome 90+, Firefox 85+, Edge 90+, Safari 14+
NFR-010	Portability	Deployable on AWS, GCP, Azure, or on-premises Linux servers
NFR-011	Interoperability	RESTful APIs with JSON format; OpenAPI 3.0 documentation
NFR-012	AI Response	AI relevance score >= 85%; No harmful or biased content generation

10. SYSTEM ARCHITECTURE

10.1 Architecture Overview

The system follows a Three-Tier Client-Server Architecture combined with a Microservices-inspired modular design. The architecture is divided into Presentation Layer, Business Logic Layer, and Data Layer, with an additional AI Integration Layer that interfaces with external LLM APIs.

LAYER	COMPONENT	TECHNOLOGY	RESPONSIBILITY
Presentation Layer	React.js Frontend	React.js 18 + Tailwind CSS	UI rendering, user interaction, state management
API Gateway	REST API Interface	Flask Python + JWT	Request routing, authentication, rate limiting
Business Logic Layer	Core Services	Flask Blueprints (Python)	Business rules, data processing, AI orchestration
AI Integration Layer	LLM API Client	Gemini API / OpenAI API	AI response generation, prompt engineering
Data Access Layer	ORM & Queries	SQLAlchemy + MySQL	Database CRUD operations, query optimization
Data Layer	Relational Database	MySQL 8.0	Persistent data storage, relational integrity
File Storage	Upload Handler	Flask FileStorage	Resume and document file management
Cache Layer	Response Cache	Redis (optional)	API response caching for performance

10.2 Component Interaction Flow

The interaction between components follows a request-response cycle. A student initiates an action on the React.js frontend, which sends an HTTP request with a JWT Bearer token to the Flask REST API. The API validates the token, processes the request through the appropriate service module, and if AI processing is needed, constructs a structured prompt and calls the Gemini/OpenAI API. The AI response is parsed, stored in MySQL if required, and returned to the frontend as a JSON response. The frontend then renders the result using React state management (useState/useContext hooks).

10.3 Deployment Architecture

The application is designed for cloud deployment on services such as AWS EC2 or Google Cloud Platform. The frontend is served via Nginx as a reverse proxy, while the Flask backend runs with Gunicorn WSGI server. MySQL database runs on a managed instance (RDS or Cloud SQL) with automated backups. SSL/TLS certificates are managed via Let's Encrypt for HTTPS enforcement.

11. DATA FLOW DIAGRAM (DFD)

11.1 Level 0 DFD (Context Diagram)

The Level 0 DFD represents the system as a single process showing the data flows between the system and external entities.

External Entity	Data Flow (Input)	Process	Data Flow (Output)	Destination
Student	Profile data, queries, resume	AI Learning System	Roadmap, feedback, recommendations	Student
Administrator	Configuration, user management	Admin Module	Reports, system status	Administrator
AI API (Gemini)	Processed prompts	AI Integration	AI-generated responses	Core System
Job Portal API	Internship listings	Data Aggregation	Filtered recommendations	Student

11.2 Level 1 DFD

The Level 1 DFD decomposes the main system into its primary subsystems, showing detailed data flows between process modules and data stores.

Process ID	Process Name	Input Data	Output Data	Data Store
P1.0	Authentication & Profile Management	Student credentials, profile data	Auth tokens, profile records	D1: User DB
P2.0	AI Learning Assistant	Student query, conversation history	AI response, session log	D2: Conversation DB
P3.0	Roadmap Generation	Student skills, goals, timeline	Personalized roadmap JSON	D3: Roadmap DB
P4.0	Resume Processing	Uploaded resume file	Parsed data, ATS score, suggestions	D4: Resume DB
P5.0	Career Advisory	Student profile, interests	Career paths, role recommendations	D5: Career DB
P6.0	Mock Interview	Domain selection, student responses	Questions, evaluation, feedback	D6: Interview DB
P7.0	Internship Matching	Student profile, preferences	Ranked internship list	D7: Internship DB
P8.0	Analytics Engine	Activity logs, progress data	Dashboard charts, metrics	D8: Analytics DB

12. USE CASE DIAGRAM

The use case diagram identifies the interactions between actors (Student, Administrator, AI System) and the system's functional use cases.

12.1 Actors

- **Student (Primary Actor):** Initiates all learning, career, and profile-related use cases
- **Administrator (Secondary Actor):** Manages system, users, and content
- **AI System (External Actor):** Provides intelligent responses and analysis

12.2 Use Case Table

UC ID	Use Case	Actor	Precondition	Postcondition
UC-01	Register Account	Student	Student completes registration form	Account created; login enabled
UC-02	Login	Student/Admin	Valid credentials entered	JWT token issued; dashboard displayed
UC-03	Update Profile	Student	Student logged in	Profile saved in database
UC-04	Chat with AI	Student	Student submits query	AI response displayed in chat
UC-05	Generate Roadmap	Student	Goal and skills provided	Personalized roadmap displayed
UC-06	Get Career Advice	Student	Student profile complete	Career paths recommended
UC-07	Upload Resume	Student	PDF/DOCX file selected	Resume parsed and stored
UC-08	Analyze Resume	Student/AI	Resume uploaded	ATS score and feedback returned
UC-09	View Skill Gaps	Student	Job role selected	Gap analysis report generated
UC-10	Start Mock Interview	Student	Domain and difficulty selected	Questions generated; interview started
UC-11	Submit Interview Answer	Student/AI	Student submits answer	AI evaluates and provides feedback
UC-12	Browse Internships	Student	Student accesses internship module	Filtered listings displayed
UC-13	View Analytics	Student	Student accesses dashboard	Charts and metrics displayed
UC-14	Manage Users	Administrator	Admin accesses user management	User list displayed; CRUD operations available
UC-15	Generate Reports	Administrator	Admin requests report	System-wide analytics report exported

13. ACTIVITY DIAGRAM

13.1 Student Learning Flow

The activity diagram below describes the complete student journey from registration to career readiness, showing decision points and parallel processes within the system.

Step	Activity/Node	Description
1	Start	Student accesses the application
2	Decision	First-time user? → Yes: Register No: Login
3	Register	Fill registration form → Verify email → Account created
4	Login	Enter credentials → JWT validation → Dashboard loaded
5	Profile Setup	Enter skills, interests, academic details, career goals
6	Decision	Profile complete? → No: Prompt completion Yes: Continue
7	Dashboard	View personalized dashboard with modules
8	Parallel Activities	
9	AI Chat	Query learning topics → AI responds with explanations
10	Generate Roadmap	Select goal → AI creates personalized roadmap
11	Resume Upload	Upload PDF/DOCX → Parse content → AI analysis
12	Career Advisor	View career recommendations → Explore pathways
13	Mock Interview	Select domain → Answer AI questions → Get feedback
14	Internship Search	Browse filtered listings → Save preferred internships
15	End Parallel	
16	View Analytics	Dashboard updates with progress metrics
17	Decision	Goals achieved? → Yes: Request certificate No: Continue learning
18	End	Session ends / Logout

13.2 Resume Analysis Activity Flow

The student uploads a resume file (PDF/DOCX). The system validates the file format and size. If valid, the file is parsed using a text extraction library (pdfplumber/python-docx). The extracted text is sent to the AI API with a structured prompt requesting ATS score, key strengths, weaknesses, missing keywords, and specific improvement suggestions. The AI response is parsed, structured into a JSON format, stored in the database, and displayed to the student on the resume analysis results page.

14. SEQUENCE DIAGRAM

14.1 AI Chat Interaction Sequence

Step	From	To	Message / Action
1	Student	React UI	Types query and clicks 'Send'
2	React UI	Flask API	POST /api/chat {message, conversation_id} + JWT header
3	Flask API	Auth Middleware	Validate JWT token → Extract user_id
4	Flask API	MySQL DB	Fetch conversation history for context
5	MySQL DB	Flask API	Return previous messages array
6	Flask API	Gemini API	POST /v1/chat with system_prompt + history + new message
7	Gemini API	Flask API	Return AI response (streaming/complete)
8	Flask API	MySQL DB	INSERT message and response into conversations table
9	Flask API	React UI	Return {response, conversation_id, timestamp}
10	React UI	Student	Display formatted AI response in chat window

14.2 Resume Analysis Sequence

Step	From	To	Message / Action
1	Student	React UI	Select PDF file and click 'Analyze Resume'
2	React UI	Flask API	POST /api/resume/analyze (multipart/form-data) + JWT
3	Flask API	File System	Save uploaded file to /uploads/{user_id}/
4	Flask API	Parser	Extract text using pdfplumber/python-docx
5	Parser	Flask API	Return raw text content
6	Flask API	AI API	Send structured prompt with resume text
7	AI API	Flask API	Return JSON: {ats_score, strengths[], weaknesses[], suggestions[]}
8	Flask API	MySQL DB	INSERT analysis results into resume_analyses table
9	Flask API	React UI	Return analysis results JSON
10	React UI	Student	Display ATS score, feedback cards, and improvement tips

15. DATABASE DESIGN

15.1 Database Overview

The system uses MySQL 8.0 as its relational database management system. The database design follows Third Normal Form (3NF) to minimize redundancy and ensure data integrity. The schema consists of 15 primary tables with well-defined relationships, indexes, and foreign key constraints.

Table Name	Purpose	Key Columns
users	Core student and admin accounts	user_id, email, password_hash, full_name, role, is_active, created_at
student_profiles	Extended student academic info	profile_id, user_id (FK), department, year, cgpa, github_url, linkedin_url
skills	Master skill reference table	skill_id, skill_name, category, proficiency_levels
student_skills	Student-to-skill mapping	id, user_id (FK), skill_id (FK), proficiency_level, verified
career_goals	Student career aspirations	goal_id, user_id (FK), target_role, target_industry, target_timeline
learning_roadmaps	AI-generated roadmaps	roadmap_id, user_id (FK), title, goal, content_json, created_at
roadmap_progress	Roadmap module completion	progress_id, roadmap_id (FK), module_index, is_completed, completed_at
conversations	AI chat history	conv_id, user_id (FK), title, created_at, last_message_at
messages	Individual chat messages	message_id, conv_id (FK), role, content, tokens_used, created_at
resumes	Uploaded resume files	resume_id, user_id (FK), file_path, file_name, uploaded_at
resume_analyses	AI resume analysis results	analysis_id, resume_id (FK), ats_score, strengths, weaknesses, suggestions, created_at
mock_interviews	Interview sessions	interview_id, user_id (FK), domain, difficulty, status, score, started_at
interview_qa	Q&A; pairs per session	qa_id, interview_id (FK), question, student_answer, ai_feedback, score
internships	Internship listings	internship_id, company, title, domain, location, duration, stipend, deadline, apply_url
saved_internships	Student saved internships	id, user_id (FK), internship_id (FK), saved_at
achievements	Badges and milestones	achievement_id, user_id (FK), badge_name, description, earned_at

16. ER DIAGRAM

The Entity-Relationship diagram defines the data model and relationships between entities in the system. Key relationships are listed below:

Entity A	Card A	Card B	Entity B	Relationship Description
users	1	1	student_profiles	One user has one extended profile
users	1	M	student_skills	One user has many skill mappings
users	1	M	conversations	One user has many chat sessions
conversations	1	M	messages	One conversation has many messages
users	1	M	learning_roadmaps	One user can have multiple roadmaps
learning_roadmaps	1	M	roadmap_progress	Each roadmap has multiple module progress records
users	1	M	resumes	One user can upload multiple resume versions
resumes	1	1	resume_analyses	Each resume has one analysis result
users	1	M	mock_interviews	One user can take multiple interview sessions
mock_interviews	1	M	interview_qa	Each session has multiple Q&A; records
users	M	M	internships	Users save many internships (via saved_internships)
skills	M	M	users	Skills linked to many users (via student_skills)

17. TECHNOLOGY STACK & JUSTIFICATION

Technology	Category	Justification
React.js 18	Frontend Framework	Component-based architecture; Virtual DOM for performance; Large ecosystem; SPA capabilities
Tailwind CSS 3	UI Styling	Utility-first CSS; Rapid prototyping; Consistent design system; No CSS bloat
Axios	HTTP Client	Promise-based; Interceptors for JWT; Clean API; Error handling support
Recharts	Data Visualization	React-native charts; Responsive; D3-powered; Easy integration
Flask 3.0 (Python)	Backend Framework	Lightweight; Flexible; Pythonic; Easy AI library integration; Blueprint modules
SQLAlchemy	ORM	Pythonic ORM; Migration support; N+1 query prevention; Secure parameterized queries
MySQL 8.0	Database	ACID compliance; Full-text search; JSON column support; Mature and reliable
Gemini API / GPT-4	AI Engine	State-of-the-art LLM; Context window support; JSON mode; Streaming; Fine-tuning options
JWT (PyJWT)	Authentication	Stateless; Secure; Expiry support; Role-based claims; Industry standard
pdfplumber	PDF Parsing	Accurate text extraction; Table detection; Layout preservation
python-docx	DOCX Parsing	Full DOCX support; Paragraph/style extraction; Microsoft format compatibility
bcrypt	Password Hashing	Adaptive hashing; Salted; Slow by design; Industry best practice
Gunicorn	WSGI Server	Production-grade; Multi-worker support; Compatible with Nginx
Nginx	Reverse Proxy	High performance; SSL termination; Static file serving; Load balancing
Docker	Containerization	Environment consistency; Easy deployment; Scalability; CI/CD integration

18. AI MODEL ARCHITECTURE

18.1 AI Integration Strategy

The system employs a Prompt Engineering-First approach, utilizing the Gemini 1.5 Pro or OpenAI GPT-4 model through their respective APIs. Rather than training custom models, the architecture focuses on sophisticated prompt design, context injection, and response parsing to achieve domain-specific AI behavior for educational use cases.

18.2 Prompt Engineering Framework

Technique	Applied In	Description
System Prompt	All modules	Defines AI persona as 'EduBot', an academic counselor. Sets tone, domain, and constraints.
Context Injection	Learning, Career	Injects student profile (skills, goals, year) as structured JSON context in every request.
History Window	Chat Module	Includes last 10 conversation turns for contextual continuity.
Structured Output	Resume, Interview	Forces JSON response format for machine-parseable structured outputs.
Few-Shot Examples	Roadmap, Skill Gap	Provides 2-3 example outputs to guide format and quality of AI responses.
Chain of Thought	Career Advice	Instructs AI to reason step-by-step before providing career recommendations.

18.3 AI Module Configurations

Module	Model	Max Tokens	Temperature	Purpose
AI Learning Assistant	gemini-1.5-pro	2048	0.7	Conversational academic Q&A;
Roadmap Generator	gemini-1.5-pro	4096	0.3	Structured JSON roadmap generation
Career Advisor	gemini-1.5-pro	2048	0.5	Career path analysis and recommendation
Resume Analyzer	gemini-1.5-pro	3000	0.2	Deterministic ATS scoring and feedback
Interview Generator	gemini-1.5-pro	1500	0.8	Creative, varied interview questions
Interview Evaluator	gemini-1.5-pro	2000	0.2	Objective answer evaluation
Skill Gap Analyzer	gemini-1.5-pro	2048	0.1	Structured gap analysis with percentages

19. MODULE-WISE FEATURE BREAKDOWN

M1: AI Learning Assistant

- Natural language question-answering for all academic subjects
- Contextual follow-up questions maintaining conversation history
- Code explanation and debugging assistance for programming subjects
- Concept simplification with analogies and examples
- Reference suggestions for further reading
- Multi-turn conversation with session persistence

M2: Personalized Learning Roadmap

- Goal-based roadmap generation (e.g., 'Become a Full Stack Developer')
- Skill-gap-aware milestone sequencing
- Resource recommendations for each roadmap module (courses, books, projects)
- Timeline estimation based on student availability
- Interactive roadmap with progress toggle per module
- Roadmap regeneration with updated skills

M3: AI Career Advisor

- Role recommendation based on skills and interests
- Industry trend analysis and job market insights
- Salary range information for recommended roles
- Career path visualization (Junior → Mid → Senior levels)
- Alternative career pathway suggestions
- Day-in-the-life descriptions for recommended roles

M4: Resume Analyzer

- PDF and DOCX resume upload and parsing
- ATS compatibility score (0-100) with breakdown
- Keyword density analysis for target job roles
- Section completeness check (Education, Experience, Skills, Projects)
- Quantification recommendations ('Increased sales by X%')
- Bullet point strength evaluation and rewrites

M5: Skill Gap Analysis

- Job role selection from 50+ predefined roles
- Side-by-side comparison of student skills vs. required skills
- Gap percentage calculation per skill category
- Priority-ranked learning recommendations to close gaps
- Time-to-bridge estimate for each skill
- Direct integration with roadmap generation module

M6: Mock Interview System

- Domain selection: Frontend, Backend, Data Science, AI/ML, DevOps, etc.
- Difficulty levels: Beginner, Intermediate, Advanced
- Question types: Technical, Behavioral, Situational, HR
- AI evaluation of student answers with scoring rubric
- Detailed feedback with model answer suggestions
- Interview session history and score trends

M7: Internship Recommendation

- Database of 500+ internship listings
- Profile-based smart matching algorithm
- Filters: Domain, Location, Duration, Stipend, Work Mode
- Save and bookmark preferred internships
- Application deadline reminders
- Direct application link integration

M8: Learning Analytics Dashboard

- Skill growth radar chart over time
- Learning hours tracked and visualized
- Roadmap completion percentage bar charts
- Interview performance trend line chart
- Career readiness score composite metric
- Weekly and monthly activity summaries

20. API DESIGN

20.1 API Design Principles

The backend exposes a RESTful API following HTTP conventions. All endpoints return JSON responses with standardized structure: {success: bool, data: object, message: string}. Authentication uses Bearer JWT tokens in the Authorization header. The API version is prefixed as /api/v1/.

Method	Endpoint	Auth	Description
POST	/api/v1/auth/register	None	Register new student account
POST	/api/v1/auth/login	None	Authenticate user; return JWT token
POST	/api/v1/auth/refresh	JWT	Refresh expired JWT token
GET	/api/v1/profile	JWT	Get current user's full profile
PUT	/api/v1/profile	JWT	Update student profile information
POST	/api/v1/chat/message	JWT	Send message to AI assistant
GET	/api/v1/chat/history/{id}	JWT	Get conversation history by ID
POST	/api/v1/roadmap/generate	JWT	Generate personalized learning roadmap
GET	/api/v1/roadmap/list	JWT	List all roadmaps for user
PUT	/api/v1/roadmap/{id}/progress	JWT	Update module completion status
POST	/api/v1/resume/upload	JWT	Upload resume file (PDF/DOCX)
POST	/api/v1/resume/analyze/{id}	JWT	Trigger AI analysis of uploaded resume
GET	/api/v1/resume/analysis/{id}	JWT	Get resume analysis results
GET	/api/v1/career/advice	JWT	Get AI career advice based on profile
POST	/api/v1/skills/gap-analysis	JWT	Get skill gap analysis for a job role
POST	/api/v1/interview/start	JWT	Start a new mock interview session
POST	/api/v1/interview/{id}/answer	JWT	Submit answer and receive AI feedback
GET	/api/v1/internships	JWT	List internships with filters
GET	/api/v1/internships/recommended	JWT	Get AI-recommended internships
GET	/api/v1/analytics/dashboard	JWT	Get analytics data for dashboard
GET	/api/v1/admin/users	Admin	List all users (admin only)
DELETE	/api/v1/admin/users/{id}	Admin	Delete user account (admin only)

21. UI/UX WIREFRAMES

The UI/UX design follows Material Design principles combined with Tailwind CSS utility classes. The interface is fully responsive and optimized for desktop and tablet screens. The following section describes the wireframe layout for each primary screen.

Dashboard Screen

- Top navigation bar with logo, user avatar, and notification bell
- Left sidebar with module icons: Chat, Roadmap, Career, Resume, Interview, Internships, Analytics
- Main content area showing: Career Readiness Score (circular progress), Recent Activity Feed, Recommended Next Steps
- Quick action cards for each module with call-to-action buttons
- Bottom statistics bar: Total Learning Hours, Modules Completed, Interviews Done

AI Chat Screen

- Full-height chat interface with message bubbles (student: right-aligned, AI: left-aligned)
- Typing indicator animation when AI is processing
- Code block rendering with syntax highlighting for programming queries
- Chat session list in left panel for conversation history
- Input bar with send button and character counter

Roadmap Screen

- Goal input form with skill self-assessment sliders
- Generated roadmap displayed as vertical timeline with connected nodes
- Each node shows: Topic name, estimated hours, resources, and completion checkbox
- Progress bar at top showing overall roadmap completion percentage
- Regenerate roadmap option with updated parameters

Resume Analyzer Screen

- Drag-and-drop file upload zone with format indicators
- Analysis loading animation during processing
- Results section: ATS Score gauge chart, Strengths list, Weaknesses list
- Detailed suggestions in expandable accordion cards
- Download analysis report button

Mock Interview Screen

- Domain and difficulty selector with visual cards
- Timer bar showing remaining session time
- Question display card with domain badge
- Text area for student answer input
- Submit button triggering AI evaluation
- Feedback panel with score, evaluation, and model answer

22. DATABASE SCHEMA

22.1 Core Tables DDL

Table: users

Column	Data Type	Constraints
user_id	INT	PRIMARY KEY AUTO_INCREMENT
email	VARCHAR(255)	UNIQUE NOT NULL
password_hash	VARCHAR(255)	NOT NULL
full_name	VARCHAR(100)	NOT NULL
role	ENUM('student','admin')	DEFAULT 'student'
is_active	BOOLEAN	DEFAULT TRUE
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP

Table: student_profiles

Column	Data Type	Constraints
profile_id	INT	PRIMARY KEY AUTO_INCREMENT
user_id	INT	FK → users(user_id) ON DELETE CASCADE
department	VARCHAR(100)	NOT NULL
year_of_study	INT	CHECK (year_of_study BETWEEN 1 AND 4)
cgpa	DECIMAL(3,2)	CHECK (cgpa BETWEEN 0 AND 10)
github_url	VARCHAR(255)	NULL
linkedin_url	VARCHAR(255)	NULL
career_goal	TEXT	NULL
bio	TEXT	NULL

Table: mock_interviews

Column	Data Type	Constraints
interview_id	INT	PRIMARY KEY AUTO_INCREMENT
user_id	INT	FK → users(user_id) ON DELETE CASCADE
domain	VARCHAR(100)	NOT NULL
difficulty	ENUM('beginner','intermediate','advanced')	NOT NULL
status	ENUM('in_progress','completed','abandoned')	DEFAULT 'in_progress'
total_score	DECIMAL(5,2)	NULL

Column	Data Type	Constraints
started_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
completed_at	TIMESTAMP	NULL

Table: resume_analyses

Column	Data Type	Constraints
analysis_id	INT	PRIMARY KEY AUTO_INCREMENT
resume_id	INT	FK → resumes(resume_id) ON DELETE CASCADE
ats_score	INT	CHECK (ats_score BETWEEN 0 AND 100)
strengths	JSON	NOT NULL
weaknesses	JSON	NOT NULL
suggestions	JSON	NOT NULL
keywords_found	JSON	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

23. PROJECT DEVELOPMENT TIMELINE (30 DAYS)

Phase	Sprint/Day	Tasks & Deliverables
Week 1 Days 1-7	Planning & Setup	Requirements finalization, tech stack setup, DB schema design, API contract definition, project scaffolding, Git repo setup
	Day 1-2	Requirements document, project setup, Flask & React initialization
	Day 3-4	MySQL schema creation, SQLAlchemy models, JWT auth implementation
	Day 5-7	User registration/login API, React auth pages, basic dashboard layout
Week 2 Days 8-14	Core AI Features	AI Learning Assistant, Roadmap Generator, Career Advisor implementation
	Day 8-9	Gemini API integration, AI chat module backend + frontend
	Day 10-11	Roadmap generation with prompt engineering and React roadmap UI
	Day 12-13	Career advisor module, skill input, and career path display
	Day 14	Integration testing of Week 2 modules
Week 3 Days 15-21	Resume & Interview Modules	Resume analyzer, skill gap analysis, mock interview system
	Day 15-16	Resume upload API, pdfplumber parsing, AI analysis endpoint
	Day 17-18	Resume analysis UI with ATS score gauge and feedback cards
	Day 19-20	Mock interview system: question generation, answer evaluation, feedback
	Day 21	Skill gap analysis module with role comparison charts
Week 4 Days 22-30	Finishing & Deployment	Internship system, analytics dashboard, testing, bug fixes, deployment
	Day 22-23	Internship database seeding, recommendation algorithm, listings UI
	Day 24-25	Analytics dashboard with Recharts visualization components
	Day 26-27	Progress tracking, achievement badges, profile completeness scoring
	Day 28	End-to-end testing, bug fixing, performance optimization
	Day 29	Docker containerization, Nginx setup, deployment to cloud
	Day 30	Final testing, documentation, project report, demo preparation

24. TESTING STRATEGY

24.1 Testing Levels

Test Type	Scope	Tool	Acceptance Criteria
Unit Testing	Individual functions and API endpoints	pytest (Backend), Jest (Frontend)	70% minimum code coverage
Integration Testing	Module-to-module and API-to-DB interactions	pytest + SQLAlchemy test DB	All API endpoints pass
System Testing	End-to-end user journeys	Selenium / Playwright	All 15 use cases pass
Performance Testing	Load and stress testing	Apache JMeter	< 2s response at 500 concurrent users
Security Testing	Vulnerability assessment	OWASP ZAP / Burp Suite	Zero critical vulnerabilities
User Acceptance Testing	Student usability and satisfaction	Manual testing with 20 students	SUS score >= 75
AI Quality Testing	AI response relevance and accuracy	Custom evaluation rubric	85% relevance on 100 test cases

24.2 Test Cases

TC ID	Test Case	Input	Expected Output	Status
TC-001	User Registration	Valid email, name, password	Account created; 201 response	Pass
TC-002	Duplicate Email	Existing email re-registration	400 error; 'Email exists' message	Pass
TC-003	JWT Login	Correct credentials	200 response with JWT token	Pass
TC-004	Invalid Login	Wrong password	401 Unauthorized response	Pass
TC-005	AI Chat Response	Send 'Explain recursion'	Relevant explanation returned	Pass
TC-006	Roadmap Generation	Goal: Full Stack Dev, 3 months	Structured roadmap JSON returned	Pass
TC-007	Resume Upload	Valid PDF file, < 5MB	File stored; resume_id returned	Pass
TC-008	Resume Analysis	Upload software engineer resume	ATS score 0-100 with feedback	Pass
TC-009	Invalid File Type	Upload .exe file	400 error; 'Invalid file type'	Pass
TC-010	Mock Interview Start	Domain: Python, Level: Intermediate	5 questions generated	Pass
TC-011	Interview Answer Eval	Submit answer to Python question	Score and feedback returned	Pass
TC-012	Skill Gap Analysis	Role: Data Scientist	Gap percentages per skill category	Pass
TC-013	Unauthorized Access	Request without JWT	401 Unauthorized	Pass

TC ID	Test Case	Input	Expected Output	Status
TC-014	SQL Injection Attempt	Inject SQL in input fields	Input sanitized; no DB error	Pass
TC-015	Analytics Dashboard	Access after 5 sessions	Charts render with correct data	Pass

25. SECURITY CONSIDERATIONS

Authentication & Authorization

- JWT tokens with 24-hour expiry; refresh token mechanism for session continuity
- Role-based access control (RBAC) distinguishing Student and Administrator roles
- bcrypt with work factor 12 for password hashing; no plaintext passwords stored
- Account lockout after 5 consecutive failed login attempts

Input Validation & Injection Prevention

- SQLAlchemy parameterized queries prevent SQL injection attacks
- Server-side input validation for all API endpoints using marshmallow schemas
- File upload validation: type whitelist (PDF, DOCX), size limit (5MB), malware scanning
- XSS prevention through React's inherent DOM sanitization and Content Security Policy headers

Data Protection

- HTTPS enforced across all endpoints using TLS 1.3
- Sensitive student data encrypted at rest using AES-256
- Database access restricted to application server IP only
- Personal data handling compliant with data protection guidelines

API Security

- Rate limiting: 100 requests/minute per user, 1000/minute per IP
- CORS policy configured to accept requests from approved frontend origins only
- API keys for Gemini/OpenAI stored in environment variables, never in codebase
- Request payload size limits to prevent DoS attacks

Logging & Monitoring

- All authentication events logged with timestamp, IP, and user agent
- Failed login attempts logged and alerted beyond threshold
- API error rates monitored with automated alerting
- Regular security audits and dependency vulnerability scanning

26. DEPLOYMENT ARCHITECTURE

26.1 Deployment Stack

Component	Technology	Configuration Details
Cloud Provider	AWS / GCP	EC2 / Compute Engine instance (t3.medium or equivalent)
Operating System	Ubuntu 22.04 LTS	Stable, well-supported Linux distribution
Web Server	Nginx 1.24	Reverse proxy, SSL termination, static file serving
WSGI Server	Gunicorn 21	4 worker processes for concurrent request handling
Database Server	MySQL 8.0	Managed RDS instance with automated backups
Containerization	Docker 24 + Compose	Separate containers for frontend, backend, database
SSL Certificate	Let's Encrypt	Free SSL/TLS certificate with auto-renewal
Domain	Custom Domain	Pointed to cloud server IP with A record
CI/CD	GitHub Actions	Automated test, build, and deploy pipeline on push to main
Monitoring	Prometheus + Grafana	Real-time metrics, alerting, and performance dashboards

26.2 Docker Compose Services

The application is containerized using Docker Compose with three primary services: **frontend** (Node.js build + Nginx static serving), **backend** (Python Flask + Gunicorn), and **database** (MySQL 8.0 with persistent volume). An optional **redis** service provides response caching. All services communicate through a private Docker network, with only Nginx exposed on public ports 80 and 443.

27. FUTURE ENHANCEMENTS

Enhancement	Description
v1.1 – Mobile Application	Develop native iOS and Android applications using React Native, extending the platform to mobile users with push notifications for learning reminders and new internship alerts.
v1.2 – Voice Interface	Integrate speech-to-text and text-to-speech capabilities for voice-based AI interaction, making the learning assistant accessible to students with visual impairments.
v1.3 – Peer Learning Network	Add a collaborative learning feature allowing students to form study groups, share roadmaps, and peer-review resumes, fostering community-based learning.
v1.4 – LMS Integration	API integrations with Moodle, Canvas, and institutional ERPs to automatically pull academic records, course completions, and grades into student profiles.
v1.5 – Video Mock Interviews	Implement webcam-based video mock interviews with AI analysis of non-verbal cues such as eye contact, posture, and expression alongside answer evaluation.
v2.0 – Custom Fine-Tuned Model	Fine-tune a domain-specific AI model on educational content, Q&A; datasets, and career guidance corpora to improve response quality beyond general LLM capabilities.
v2.1 – Employer Portal	Create a dedicated portal for companies to post internships directly, review AI-matched candidate profiles, and schedule interviews — closing the loop between students and recruiters.
v2.2 – Multilingual Support	Add support for Tamil, Hindi, and other regional Indian languages to make the platform accessible to a broader student demographic.
v2.3 – Blockchain Credentials	Issue blockchain-verified digital certificates for completed roadmaps and mock interview achievements, providing tamper-proof proof of skills.
v2.4 – AI Mentor Matching	Match students with industry mentors based on career goals and skill sets, enabling structured mentorship programs within the platform.

28. EXPECTED OUTCOMES

28.1 Quantitative Outcomes

Outcome Metric	Expected Target
Placement Rate Improvement	15-25% improvement in department placement rates within one academic year of deployment
Resume Quality Score	Average student ATS score improving from 45 to 75+ after using the resume analyzer module
Interview Confidence	80% of students reporting increased interview confidence after 5+ mock interview sessions
Skill Gap Reduction	Average skill gap for target roles reduced by 40% within 3 months of roadmap follow-through
User Engagement	Average session duration of 25+ minutes; 4+ sessions per week per active student
Student Satisfaction	Net Promoter Score (NPS) >= 50; SUS usability score >= 75 / 100
AI Response Quality	85%+ relevance rating on AI responses in user satisfaction surveys
Internship Applications	60% of active users apply to at least one recommended internship within 30 days

28.2 Qualitative Outcomes

- Establishment of a data-driven culture of continuous self-improvement among students
- Reduction in the cognitive burden of career planning through AI-assisted guidance
- Democratization of quality career counseling, previously available only to privileged students
- Creation of a feedback loop between student performance data and curriculum improvement
- Increased awareness of industry trends and evolving skill requirements
- Building of a student community around shared learning goals and progress
- Institutional recognition as a technologically advanced department with innovative student support

29. CONCLUSION

The **AI Powered Student Learning Assistant & Personalized Career and Education Advisor** represents a significant advancement in the application of Artificial Intelligence to educational technology. By integrating ten specialized AI-driven modules into a unified, accessible platform, this project addresses the critical gap between academic learning and professional readiness that affects millions of engineering students annually.

The system's architecture — built on React.js, Flask Python, MySQL, and state-of-the-art LLM APIs — provides a robust, scalable, and secure foundation capable of serving hundreds of concurrent users while delivering highly personalized experiences. The emphasis on prompt engineering, contextual AI interaction, and data-driven analytics ensures that the system provides genuinely valuable guidance rather than generic, impersonal responses.

This project demonstrates the practical applicability of final-year engineering knowledge across full-stack development, AI/ML integration, database design, security engineering, and user experience design. The 30-day development timeline, structured as agile sprints, showcases effective project management and technical execution within academic constraints.

The expected outcomes — including improved placement rates, enhanced resume quality, increased interview confidence, and measurable skill gap reduction — underscore the real-world impact potential of this platform. Future enhancements including mobile applications, fine-tuned AI models, and employer portals position this project as a foundation for a comprehensive career development ecosystem.

In conclusion, this project not only fulfills the academic requirements of a final-year engineering major project but also creates genuine value for students, educational institutions, and the broader talent ecosystem. The convergence of AI, education, and career development in this platform exemplifies the transformative potential of AI/ML engineering in solving real-world human challenges.

Student Signature	Guide Signature	HOD Signature
Hari Priyan V	Dr. [Faculty Name]	Prof. [HOD Name]
<i>Third Year B.E. (AI&ML;)</i>	<i>Professor, Dept. of AI & ML</i>	<i>Head of Department, AI & ML</i>

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